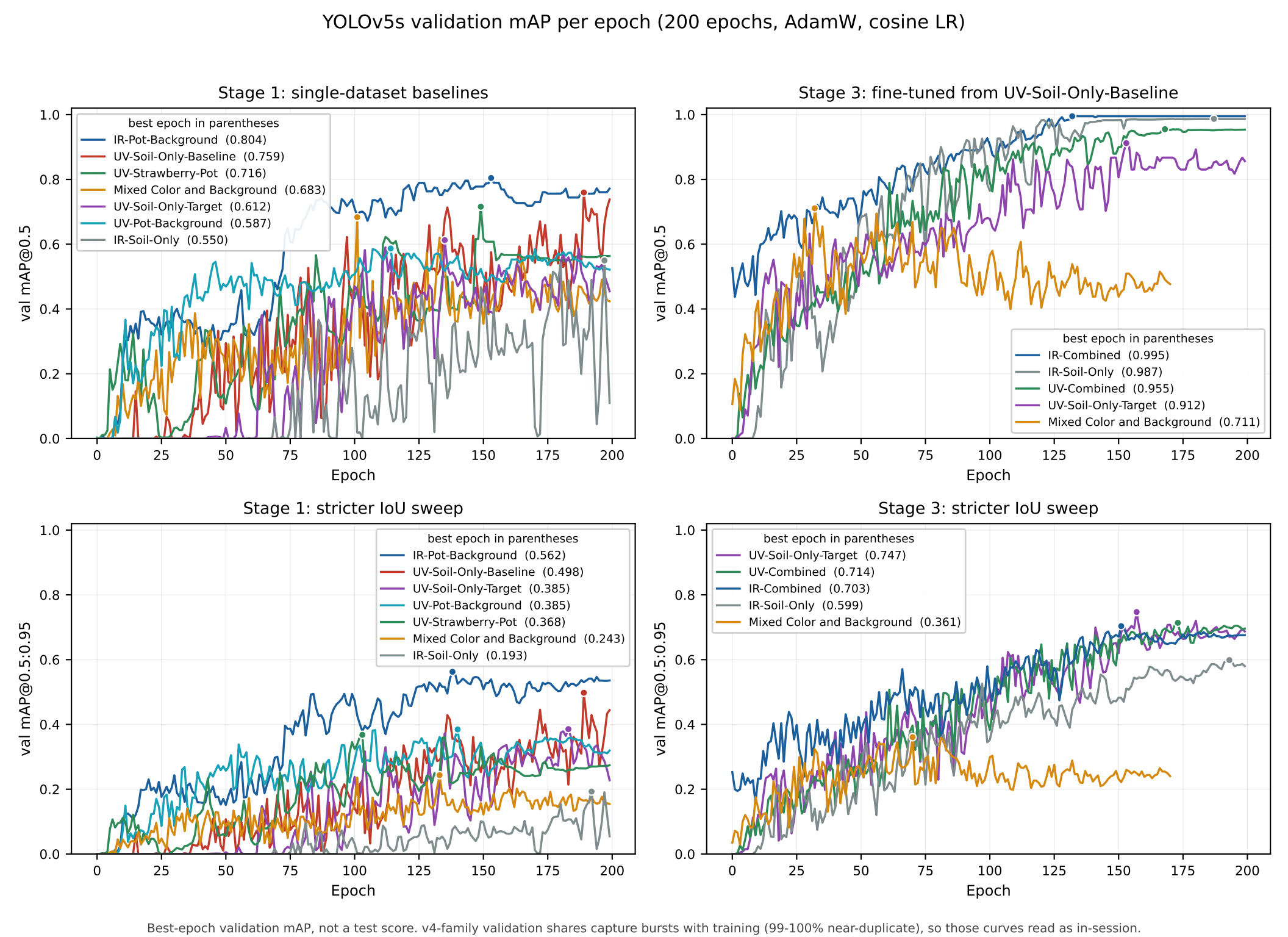

YOLOv5s validation mAP per epoch (200 epochs, AdamW, cosine LR)
Stage 1: single-dataset baselines
best epoch in parentheses
IR-Pot-Background (0.804)
UV-Soil-Only-Baseline (0.759)
UV-Strawberry-Pot (0.716)
Mixed Color and Background (0.683)
UV-Soil-Only-Target (0.612)
UV-Pot-Background (0.587)
IR-Soil-Only (0.550)
val mAP@0.5
Epoch
Stage 3: fine-tuned from UV-Soil-Only-Baseline
best epoch in parentheses
IR-Combined (0.995)
IR-Soil-Only (0.987)
UV-Combined (0.955)
UV-Soil-Only-Target (0.912)
Mixed Color and Background (0.711)
val mAP@0.5
Epoch
Stage 1: stricter IoU sweep
best epoch in parentheses
IR-Pot-Background (0.562)
UV-Soil-Only-Baseline (0.498)
UV-Soil-Only-Target (0.385)
UV-Pot-Background (0.385)
UV-Strawberry-Pot (0.368)
Mixed Color and Background (0.243)
IR-Soil-Only (0.193)
val mAP@0.5:0.95
Epoch
Stage 3: stricter IoU sweep
best epoch in parentheses
UV-Soil-Only-Target (0.747)
UV-Combined (0.714)
IR-Combined (0.703)
IR-Soil-Only (0.599)
Mixed Color and Background (0.361)
val mAP@0.5:0.95
Epoch
Best-epoch validation mAP, not a test score. v4-family validation shares capture bursts with training (99-100% near-duplicate), so those curves read as in-session.